Improving Digital Literacy of Rural Women Entrepreneurs in Indonesia

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***Abstract*—Digital technology plays a pivotal role in business development of women entrepreneurs today. It enables women entrepreneurs to have more active participation in economic sector. Nevertheless, less women have digital literacy and skill that making them difficult to expand business market. This research aims to portray women economic empowerment activities to improve rural women entrepreneurs' literacy and skill in digital usage. This research employs observation, in-depth interview and survey to 78 rural women entrepreneurs from 3 provinces in Indonesia. These women are participants of economic empowerment program conducted by Social Trust Fund of UIN Jakarta. The finding revealed that proper strategy, program compatibility and taking situation of rural women as consideration will make empowerment program becoming more optimal to improve women digital literacy.**

Keywords—Digital technology, literacy, rural women entrepreneurs, economic empowerment, training

I. Introduction

Digital technology plays a pivotal role in business development of women entrepreneurs today. Digital technology enables women entrepreneurs to have more active participation in economic sector. Despite the widespread use of technology, women are still underdeveloped. Data shows that gender gap in digital world has put women in a disadvantageous position [1]. Women especially in rural areas have been identified as disadvantaged with lower levels of education as well as fewer training opportunities [2]. Women illiteracy rate (13.85%) is twice as high as men illiteracy rate (6.26%) and this statistic worsens in rural areas. By the emerging growth of e-commerce and digital marketing today need for upskilling women with digital literacy as education has been recognized as a determinant of earnings and future economic prosperity[3].

Women's economic empowerment program has been used to help women improve their capacity. It has been conducted by both the government and non-government organizations, however, there is plenty of room for improvement [4]. Although many economic empowerment programs for women have been carried out, they have not yet reached all women SMEs actors, who account for 65% of the total SMEs actors, and some of them are located in rural. Furthermore, the quality of program implementation technically has not taken much into account the situation of women SMEs who generally play a dual role as housewives and business actors.

In general, women SMEs still experience cultural barriers which become the basis for placing women's roles. The presence of digital technology is considered to help women negotiate these barriers so that women SMEs can still take part optimally in the economic sector [5]. Unfortunately, women's digital literacy and skills, especially in rural areas, are still low. Rural SMEs especially women are less likely to access digital platforms such as social media for business purposes [6], they mostly use it for private communication, casual interaction or social friendship[7]. Though there are many women empowerment program conducted training in digital to improve women's literacy and skills, the quality and their effectiveness still needs to be evaluated as well as their sustainability are aspects that also need to be improved.

Indeed, research on women's empowerment and digital technology has been discussed on many in different themes. Among others are about urgency of digital literacy for women's empowerment[8], improving women's gender awareness through digital technology and access to information [9], and the role, opportunities and challenges to promote and implement digital literacy to women [10]. However, limited research has been conducted to study strategies applied in women's empowerment programs to increase women's digital skills effectively. Furthermore, most research on women entrepreneurs and digital technology have an urban focus[11], while, rural women entrepreneurship has different characteristics that requires analyses different from the urban context. This paper aims to portray efforts to improve the digital literacy and skills of rural women entrepreneurs through women's economic empowerment program. To what extent strategies to improve digital literacy are considered to be effective and by the conditions of women in rural areas?

II. Literature Review

A. Rural Women Entrepreneurship

In recent times, rural entrepreneurship has increasingly been taken into account because it has a potential impact on employment, food security, economic development, and social security[7]. The establishment of rural entrepreneurship implies the emergence of rural industrialization[12]-[15]. Rural entrepreneurship is not only seen from the perspective of opportunity-based but also from resource-based view[11]. Most rural entrepreneurs scale is small-scale businesses, small traders and artisans. They represent an informal sector of the economy[7]. Many of the actors of rural entrepreneurs are women, and they usually become entrepreneurs after marriage[16]. They evolved in agriculture where women usually involved in helping men and contributing to family income through on-farm and off-farm diversification[16].

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Rural women's entrepreneurship faces social systems and norms that sometimes see women as subordinate. So, they are not taken into account in the economic field. women's economic activity is only a complement to increasing family income. Most women's economic activities are a continuation of their domestic tasks and their daily lives[17]. In other name is viewed as an embedded economy[18]. In addition, rural women usually have a low level of education and limited exposure to entrepreneurial or basic business practice. Their way of managing business is more from experience and trial and error.

B. Women Economic Empowerment

Women's economic empowerment has become an imperative component to promote gender equality, improve women's capacity, reduce poverty and achieve sustainable development[19]. Women's economic empowerment program aims to improve women's ability to access resources, such as education, healthcare as well as property so they can advance their well-being and reduce poverty [20]. In Indonesia, many implementations of economic empowerment programs have been aimed at women. Economic empowerment programs that have been carried out by both the central and regional governments are the Takesra – Kukesra Program; Business Assistance (Hub) in collaboration with the Australian government through the MAMPU program, business digitization training, MSME capacity building training, Rural Area Superior Products (Prukades), Village-Owned Enterprises (BUMDes), Community-Based Enterprises (UBK), 100,000 new entrepreneurs, ABCGM Synergy, One Village One Company (OVOC), One Village One Product (OVOP), One Islamic Boarding School One Product (OPOP), Lorong Business Entity (BULo), Migrant Worker Family Community (KKBM). In addition, empowerment programs that aim at increasing the use of technology are developed by information technology companies such as CSR Companies, marketplaces and GoUKM.id. There are also empowerment programs that are carried out by NGOs, community organizations, SMEs, universities and Islamic philanthropic institutions.

In the advance of information and technology today, the ability of digital literacy is urgent for women[8]. One of women empowerment activities are to increase women's digital literacy. Digital literacy and skills have been become important components in empowering women.

C. Women's Digital Literacy

There are at least three opportunities provided by digital technology related to women's empowerment: 1) technology allows women to break down social and cultural barriers that impede women's mobility; 2) technology can improve women's social skills which are generally considered to be better than men's; 3) technology can significantly improve women's social skills through increasing cognitive abilities and digital literacy [21]. Thus, gender inequality in current economic activities can be resolved by digital technology.

OECD indicates a digital gap between men and women by showing that women have failed to reach the same level as men in digital literacy. In 2015, the number of women who had cellular phones was lower than men. In addition, women used cellular phones to perform tasks that were less complicated than men. In 2016, the number of women who went online was below 250 million which was lower than the number of men [1]. Perceptions of the use of technology in women have prevented them from exploring technology extensively in some countries. One of these perceptions is that technology has a negative impact on women. Another challenge is women's ability to own devices [1], [21].

In Indonesia, the World Wide Web Foundation's Women Rights Online discovered that only 20% of Indonesian women have internet access. Some of the reasons are geographical limitations, limited ability of women to operate devices, low education, and cultural boundaries[8]. Another issue is women have limited free time due to roles that they must play in the household[22].

In terms of women's economic empowerment, digital literacy can stimulate women's participation in economic development. Previous studies have demonstrated that technology enables women to play a more active role in social, educational, and economic activities. It also means that women's empowerment becomes more possible and can be accelerated.

In many cases, women's involvement in economic activities is restricted by cultural elements which eventually restrict women's mobility. To overcome this issue, digital technology becomes a solution for women to be able to break down cultural boundaries and barriers. Technology allows women to manage work hours and practices, and provides access to a larger market[1]. When women cannot meet with consumers because cultural boundaries do not allow them, technology has paved the way for them to meet consumers online without having to leave their homes to promote their products. So that, women can easily move between roles; both as housewives and providers for their family[22].

Women generally can contribute to economic development. Currently, many women had have successfully developed online businesses and even startup companies. The internet provides knowledge for women on how to develop their businesses and create innovation. It also accelerates the emergence of new ideas and new business opportunities through online social networks and e-commerce[23]. In addition, digital literacy also makes way for women to be involved in activities and struggles for equality. Digital technology has also triggered women to become agents for their personal development [24]. In other words, technology provides the power to fight against systemic violence and discrimination [24].

Effort to improve women's digital literacy has been conducted by many. Most of the training in digital for entrepreneurs has been provided by the local Office of Cooperatives and SMEs. Some of them have obtained it from banks, such as BRI. There are also trainings provided by e-commerce companies in collaboration with the local government or with local NGO. On example is the training provided by Blibli.com in collaboration with PLUT (Department of Cooperatives and SMEs).

III. Methodology

A. Research Approach

The research used qualitative approach. The object of this research is women entrepreneurs in rural area who become participants of women economic empowerment program conducted by Social Trust Fund (STF) UIN Jakarta in 3 (three) Provinces (West Java, South Sulawesi, and South East Nusa) in Indonesia.

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B. Data Source and Collection Procedures

This research uses primary data collected through surveys and in-depth interviews and observation. There are 78 women entrepreneurs becoming respondents. They are coming from 3 provinces. 43% is coming from South East Nusa (Nusa Tenggara Barat), 31% is from Sulawesi, and 26% is from West Java. The women entrepreneurs in South East Nusa are coming from Lombok Tengah and Lombok Timur. In South Sulawesi, there are coming from Enrekang and Bantaeng. In West Java, the women are from Cianjur and Tasikmalaya.

IV. Result and Discussion

A. Demographic Analysis

A number of women entrepreneurs who become a sample in this study is 78. The majority of them are married with ages is ranging between 30 to 50 years old. 21% of them hold bachelor's degree, and 50% graduated from Senior high school. The rest are graduated from Junior high school and elementary school constituting 19% and 10% respectively. The women do business that relates to their daily routine. 63% of them do business in food, 19% in craft, 10% in hospitality and only 8% in fashion (Table 1).

TABLE I. DEMOGRAPHY OF WOMEN ENTREPRENEUR

	Criteria		%
1	Marital Status	Married	85%
		Not Married	6%
		Widow	9%
2	Ages	20-30	8%
		> 30-40	38%
		> 40-50	44%
		> 50	10%
4	Level of Education	Elementary school	10%
		Junior high school	19%
		Senior high school	50%
		Bachelor Degree	21%
5	Type of products	Food	63%
		Craft	19%
		Hospitality	10%
		Fashion	8%

B. Digital Platform used by rural women entrepreneurs

In general, women in rural know already about various digital platforms, especially social media. Among all the social media, WhatsApp and Facebook are the most popular. The majority of women (86%) use WhatsApp. Facebook is in the second position. The third position most used is Instagram, while Marketplace and Google My Business are less popular among rural women entrepreneurs (Table 2).

TABLE II. DIGITAL PLATFORM USED BY WOMEN ENTREPRENEURS

No	Digital Platform	The way women operate the platform		
		Not using	Operate by assistant	Self-operate
1	WhatsApp	5%	9%	86%
2	Facebook	22%	8%	70%
3	Instagram	32%	12%	56%
4	Marketplace	91%	3%	6%
5	Google My Business	56%	23%	21%

C. Women illiteracy rate

The above information about the use of digital platforms by rural women is confirmed by women's capacity to use digital platforms. Among 78 women 45% of them admitted they less understood in using digital platform for business. 36% of women has average capacity in using digital's platform, and the rest, 19% have good capacity in using digital's platform. These data could indicate illiteracy rates among women in rural area. The number of women illiteracy rates by this research is quite high.

D. Digital Training and Mentoring as Strategy for Digital Literacy Improvement

The program of economic empowerment conducted by STF consists of two main activities; training on digital marketing and mentoring. The strategy used to conduct the training and mentoring is by doing collaboration with the government and the university. Collaboration with the government intended to find synergy and support from the government. In addition to, rural people have high respect to the government. They will welcome any program coming from the government. Collaboration with the university aims to involve students as assistants to accompany the participants in learning and applying digital. university students who were currently conducting community service (KKN) in three provinces.

First is training, this study found that training to introduce digital technology for women entrepreneurs in rural areas has not been carried out much. Many of the women in this research admitted that they had not received or never attended the digital training beforehand. As admitted by women in Enrekang, South Sulawesi. None of them attended digital training. In Bantaeng only 20% of the women have attended training. In Cianjur, only 40% of admitted have been trained. The same condition also occurs in Lombok. Even if they have, on average, they have only attended one to two training sessions.

This study found that even though the women already use social media, not all of them are capable of operating the platforms by themselves. They need other people to help. But many of them can independently operate the platform. This independence shows that the two applications make it easy for women to understand how they work (see Table 2).

Even though they can use and operate the platform by themselves, a limited of them can create their own social media account. Many of the women do not understand the process of creating an account including creating an email account as a requirement to make a social account or other digital platform. So, the training conducted by STF has to teach the women the very basics of knowledge on social media. The training program targeted women to market their products in the digital market. the training was conducted in several sessions. They are; introducing women to digital platform for business, and how to present products in social media (training on photography techniques and how to write interesting content marketing).

Second is the mentoring. Mentoring plays a vital part in the success of the training. Mentoring is also a way to evaluate how well the participants understand the training material. The formal mentoring was provided by students and consultants from the Cooperative-Micro-Small-Medium Business Integrated Service Center (PLUT-KUMKM). Students provided mentoring for MSMEs owners and observed their

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progress in using digital. Along with the community service program, students can live in the area and provide direct mentoring to women who own MSMEs for one month. Through this mentoring, students can evaluate their understanding of the material and their ability to apply the material.

Through the community service program (KKN), digital mentoring activities can be carried out intensively because students live close to the entrepreneurs. Moreover, women entrepreneurs, who are also a mother and wife, are preoccupied with family and production activities. Considering the amount of time that these women have, the activity schedule is set to be flexible. In mentoring activities, students strengthened the digital skills for female MSMEs owners and helped them to solve some problems that they encountered when they applied digital marketing.

Furthermore, the mentoring was continued by Consultants from the Cooperative-Micro-Small-Medium Business Service Center (PLUT-KUMKM) in three provinces. To ensure that MSMEs entrepreneurs are able to independently promote their products digitally, the consultants visited the MSMEs locations for four months.

Women admit that the training and mentoring have been supporting their product marketing and sales. They feel that after attending the training, their sales and income have increased. Moreover, they are more confident in creating marketing narratives for their products on social media and encouraging them to sell other products.

E. Issues in Training and Mentoring

Most women found the process of creating an account on the Marketplace was 'complicated' (impractical). They felt that too much time was spent on creating emails, filling out identity forms and uploading many things to use the marketplace. Through direct mentoring from the PLUT consultant, women are now able to make Call To Action (CTA) feature buttons. They are also able to activate messaging features, share products, and create profiles and URLs (Uniform Resource Locator) or links that connect from Google My Business to WhatsApp in the format wa.me/6281xxx. Moreover, entrepreneurs can understand and use the offer posting feature on Google My Business.

Informal mentoring provided by family members is also very important for the success of this program. Women usually ask for help from family members such as children or nephews to operate their social media. On the one hand, the mentoring provided mainly by family members has many benefits for the participants. On the other hand, it could boomerang on them. For example, most women have Marketplace accounts and passwords email created by their children or nieces. When they are going to use their accounts again after a long time, they forget the password of their accounts. This incident frequently occurs in Enrekang and Bantaeng. Another interesting incident is that many women refuse to register their email with their date of birth. Some women do not want to share their personal information, and other women feel they have forgotten because they never recorded it. Instead, they write the date of birth of their child who is a toddler. As a result, the process of creating an email always fails. Actually, one of the requirements for creating an email account is that the applicant must be at least 13 years old.

In terms of technology usage, attitudes of inferiority often occur among women. They feel that the technology is not suitable for their age and they lack of creativity in advertising. Therefore, the urge to maximize technology as a means of doing business is not very visible. This study believes that a "slow" (relaxed) mentoring strategy is the right strategy. This strategy tries to adjust the time that women need and understand their needs regarding the use of technology. The mentorship which is provided by Community Service program (KKN) students adopts a "slow" strategy. They are asked to remain patient during the training to ensure that women MSME owners fully understand how to use digital applications such as Google My Business, social media and the Marketplace. Another mentoring strategy is to bring women together with their business colleagues outside the region through WhatsApp groups and training meetings. They are also encouraged to use video calling and Zoom meeting applications independently.

V. Conclusion

Women's economic empowerment as an approach to improve women's capacity should provide a proper strategy for women in rural. The success of the women's economic empowerment program is highly dependent on the compatibility of the program with the participants (SME owners). The strategy of combining formal training activities with mentoring (private assistances) makes it easier for women entrepreneurs in villages to understand the skills of using digital platforms. Collaboration with various stakeholders makes program implementation more optimal, such as with government, university (students) and other local partners.

Furthermore, economic empowerment programs for village women do not only stop at increasing women's literacy and capacity in using digital, but also until their businesses increase or generate profits. This research also recommends that there is a government policy to improve digital infrastructure such as internet access, online sales infrastructure such as product delivery expeditions must be improved, and expanded collaboration with large entrepreneurs.

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